

## Class Graphs Code

//Students: Kobi Bourne

// Sarah Crotty

//Purpose: Class to store graph and its methods

public class Graphs

{

    //declares and initializes Array of site names

    String[] sites = {"The Lords Woods", "High Park", "Rathwood", "Mullaghreelan Woods",  
"Glendalough", "Sally's Gap", "The Wicklow Way"};

    //declares Adjacency matrix(2-D Array) storing distances

    double[][] matrix =

    {

        {0, 3.2, 2.8, 0, 0, 0, 0},

        {0, 0, 1.4, 0, 0, 0, 0},

        {0, 1.4, 0, 3.2, 3.2, 0, 0},

        {0, 0, 0, 0, 2, 5, 0},

        {0, 0, 3.2, 2, 0, 3.6, 0},

        {0, 0, 0, 0, 0, 0, 1.4},

        {0, 0, 0, 0, 0, 1.4, 0}

    };

    //method to get index of site

    private int getIndex(String site)

    {

        for(int index = 0; index < sites.length; index++) //loops through all sites

        {

            if(sites[index].equalsIgnoreCase(site)) //if site = site at current index

```

        {
            return index; //return site at index
        }
    }
    return -1; //if not found return -1
}

```

//method to print all leaving connectings for site

```
public void Search(String site)
```

```

{
    int index;
    index = getIndex(site); //get index of site calling method

```

```

    if(index == -1) //if site not found

```

```

    {
        return; //exit method
    }

```

```

    System.out.println(site + " -->"); // print site

```

```

    for(int col = 0; col < sites.length; col++) //loops through cloumns in sites row

```

```

    {
        if(matrix[index][col] !=0) //if connection
        {
            System.out.println(sites[col] + " ( " + matrix[index][col] + " ) "); //print site and
distance
        }
    }
}

```

```

        System.out.println(); // line for spacing

    }

    //add connection
    public void Insert(String site1, String site2, double weight)
    {
        int row;
        int col;

        row = getIndex(site1); //get site1 row
        col = getIndex(site2); //get site2 col

        if(row != -1 && col != -1)
        {
            matrix[row][col] = weight; //add connection, store distance at row,col
        }

        System.out.println("Added new connection");
        System.out.println(); //line for spacing
    }

    //print all connected sites
    public void AllCons(String site)
    {
        int index;

        index = getIndex(site); //get index of site calling method

        if(index == -1) //if site not found
    }

```

```
{  
    return; //exit method  
}
```

```
System.out.println(site + " Connections -->"); // print site
```

```
for(int col = 0; col < sites.length; col++) //loops through cloumns in sites row
```

```
{  
    if(matrix[index][col] !=0) //if connection  
    {  
        System.out.println(sites[col]); //print connected sites  
    }  
}
```

```
System.out.println(); // spacing
```

```
}
```

```
//print closest site
```

```
public void Closest(String site)
```

```
{
```

```
    int index;
```

```
    int nearest;
```

```
    double min;
```

```
    index = getIndex(site); //get index of site calling method
```

```
    if(index == -1) //if site not found
```

```
    {  
        return; //exit method
```

```

    }

    nearest = -1; //closest site index
    min = -1; //smallest distance so far

    for(int col = 0; col < sites.length; col++) //loops through cloumns
    {
        if(matrix[index][col] !=0) //if connection
        {
            if(min == -1 || matrix[index][col] < min) //if first or smaller
            {
                min = matrix[index][col]; //updates min
                nearest = col; //update nearest
            }
        }
    }

    if(nearest != -1) //if nearest not =-1
    {
        System.out.println("The closest site to " + site + " is " + sites[nearest]); //print
nearest site
    }

}

}

```

## Class GraphDriver Code

//Students: Kobi Bourne

// Sarah Crotty

//Purpose: Driver to show functionality

public class GraphDriver

{

public static void main(String [] args)

{

Graphs walks = new Graphs(); //new graph object

walks.Search("The Lords Woods"); //prints connections from the lords woods

walks.AllCons("Rathwood"); //prints all sites connected to rathwood

walks.Closest("Sally's Gap"); //prints closest to sally's gap

walks.Insert("The Lords Woods", "Mullaghreelan Woods", 3.3); //adds connection

walks.Search("The Lords Woods"); //check added connection

}

}